

2.2 Limit of a Function and Limit Laws

In Section 2.1 we saw how limits arise when finding the instantaneous rate of change of a function or the tangent line to a curve. We begin this section by presenting an informal definition of the limit of a function. We then describe laws that capture the behavior of limits. These laws enable us to quickly compute limits for a variety of functions, including polynomials and rational functions. We present the precise definition of a limit in the next section.

① Limits of Function Values

Frequently when studying a function $y = f(x)$, we find ourselves interested in the function's behavior *near* a particular point c , but not *at* c itself. An important example occurs when the process of trying to evaluate a function at c leads to division by zero, which is undefined. We encountered this when seeking the instantaneous rate of change in y by considering the quotient function $\frac{\Delta y}{h}$ for h closer and closer to zero. In the next example we explore numerically how a function behaves near a particular point at which we cannot directly evaluate the function.

Example 1 How does the function

$$f(x) = \frac{x^2 - 1}{x - 1}$$

behave near $x = 1$?

Solution $f(x) = \frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1, \quad x \neq 1.$

x	0.9	0.99	0.999	0.9999	$\rightarrow$	1	$\leftarrow$	1.0001	1.001	1.01	1.1
$f(x) = x + 1$	1.9	1.99	1.999	1.9999	$\rightarrow$	2	$\leftarrow$	2.0001	2.001	2.01	2.1

So, when x goes to 1 from both sides of 1, $f(x)$ gets closer and closer to 2.

We now give an informal definition of the limit of a function f at an interior point of the domain of f .

Suppose $f(x)$ is defined on

$$(c - \rho, c) \cup (c, c + \rho), \rho > 0.$$

When x approaches c ($x \neq c$) from both sides of c , if $f(x)$ gets closer and closer to L , we say that f has the limit L as x approaches c , and write

$$\lim_{x \rightarrow c} f(x) = L.$$

Our definition here is “informal” because phrases like close and close are imprecise. We will need the precise definition given in Section 2.3.

Thus, from Example 1, we have

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

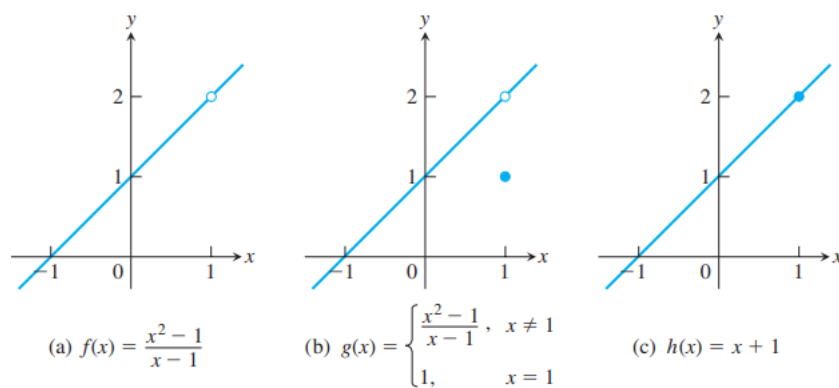
Note 1 The limit value of a function does not depend on how the function is defined at the point being approached.

For example, $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} h(x) = 2$, where

$$(a) f(x) = \frac{x^2 - 1}{x - 1},$$

$$(b) g(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1; \\ 1, & x = 1. \end{cases},$$

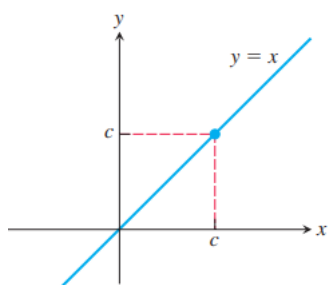
$$(c) h(x) = x + 1.$$



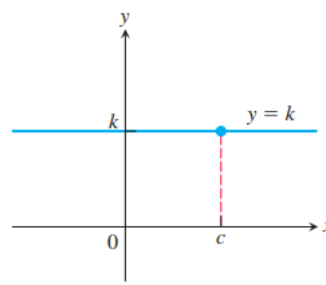
Note 2 If we know the graph of a function, we can easily determine the limit of the function.

Example 2

- (a) If $f(x) = x$, then $\lim_{x \rightarrow c} f(x) = c$.
- (b) If $f(x) = k$, and k is a constant, then $\lim_{x \rightarrow c} f(x) = k$.



(a) Identity function



(b) Constant function

Example 3 Discuss the behavior of the following functions as $x \rightarrow 0$, explaining why they have no limit as $x \rightarrow 0$.

a) $u(x) = \begin{cases} 0, & x < 0; \\ 1, & x \geq 0. \end{cases}$

$$\text{b) } g(x) = \begin{cases} \frac{1}{x}, & x \neq 0; \\ 0, & x = 0. \end{cases}$$

$$\text{c) } f(x) = \begin{cases} 0, & x \leq 0; \\ \sin \frac{1}{x}, & x > 0. \end{cases}$$

② The Limit Laws

THEOREM 1—Limit Laws If L , M , c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M, \quad \text{then}$$

1. *Sum Rule:* $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$
2. *Difference Rule:* $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$
3. *Constant Multiple Rule:* $\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$
4. *Product Rule:* $\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$
5. *Quotient Rule:* $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$
6. *Power Rule:* $\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \text{ a positive integer}$
7. *Root Rule:* $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$

(If n is even, we assume that $\lim_{x \rightarrow c} f(x) = L > 0$.)

Example 4 Use the observations $\lim_{x \rightarrow c} x = c$ and $\lim_{x \rightarrow c} k = k$ (Example 2) and the

fundamental rules of limits to find the following limits.

$$\text{(a) } \lim_{x \rightarrow c} (x^3 + 4x^2 - 3)$$

$$\text{(b) } \lim_{x \rightarrow c} \frac{x^4 + x^2 - 1}{x^2 + 5}$$

$$\text{(c) } \lim_{x \rightarrow c} \sqrt{4x^2 - 3}$$

THEOREM 2—Limits of Polynomials

If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$, then

$$\lim_{x \rightarrow c} P(x) = P(c) = a_n c^n + a_{n-1} c^{n-1} + \cdots + a_0.$$

THEOREM 3—Limits of Rational Functions

If $P(x)$ and $Q(x)$ are polynomials and $Q(c) \neq 0$, then

$$\lim_{x \rightarrow c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}.$$

Example 5 Find the following limits.

(a) $\lim_{x \rightarrow c} (x^3 + 4x^2 - 3)$

(b) $\lim_{x \rightarrow -1} \frac{x^3 + 4x^2 - 1}{x^2 + 5}$

(c) $\lim_{x \rightarrow -1} \frac{2x^2 - x + 5}{x^2 + 3x + 1}$

③ Eliminating Common Factors from Zero Denominators

Theorem 3 applies only if the denominator of the rational function is not zero at the limit point c . If the denominator is zero, canceling common factors in the numerator and denominator may reduce the fraction to one whose denominator is no longer zero at c . If this happens, we can find the limit by substitution in the simplified fraction.

Example 6 Evaluate

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}.$$

Example 7 Evaluate

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}.$$

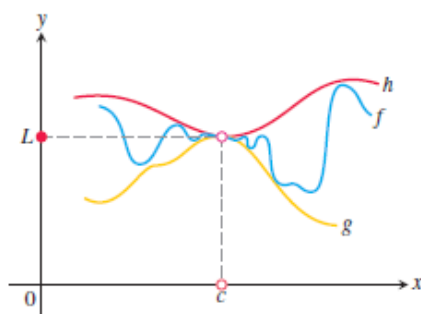
Example 8 Evaluate

$$\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}.$$

The next theorem shows how to evaluate difficult limits by comparing them with functions having known limits.

④ The Sandwich Theorem

The following theorem enables us to calculate a variety of limits. It is called the Sandwich Theorem because it refers to a function f whose values are sandwiched between the values of two other functions g and h that have the same limit L at a point c . Being trapped between the values of two functions that approach L , the values of f must also approach L .



THEOREM 4—The Sandwich Theorem Suppose that $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing c , except possibly at $x = c$ itself. Suppose also that

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L.$$

Then $\lim_{x \rightarrow c} f(x) = L$.

Example 9 Given that

$$1 - \frac{x^2}{4} \leq u(x) \leq 1 + \frac{x^2}{2}$$

for all $x \neq 0$, find $\lim_{x \rightarrow 0} u(x)$, no matter how complicated u is.

Example 10 Show that

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0.$$

Finally, we give an example from homework.

Example 11 For the function $f(x)$ graphed here, find the following limits or explain why they do not exist.

- (a) $\lim_{t \rightarrow -2} f(t)$
- (b) $\lim_{t \rightarrow -1} f(t)$
- (c) $\lim_{t \rightarrow 0} f(t)$
- (d) $\lim_{t \rightarrow -0.5} f(t)$

